

FPGA-Based Defect Detection in PCBs Using Image Processing

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Abstract – Efficient and accurate PCB defect detection relies on robust image processing. This paper presents for Printed Circuit Boards (PCBs) during the fabrication process. The primary objective is to develop an inexpensive and comprehensive defect detection system utilizing FPGA-based image processing, specifically binary image analysis and pixel-wise comparison, to achieve real-time defect localization. This system eliminates the subjective aspects of manual inspection while providing real-time defect identification and localization. The proposed technique uses a digital image of an ideal PCB as a reference, against which other PCBs are analyzed. The process begins by acquiring PCB images and converting them into grayscale and binary formats. These images are then compared using Verilog-based row and column array analysis on an FPGA, enabling the detection of defects such as open circuits, short circuits, and missing or misaligned components. The system identifies defects based on pixel-wise comparisons and determines their exact locations in a 1024x1024 resolution grid. The results are processed and displayed to facilitate precise evaluation of the PCB's quality.

Key Words: FPGA, Verilog, FPGA-accelerated image processing, Defect Detection, Binary image analysis, PCB Inspection, Automation, Real-time image analysis, Hardware-based image processing.

I. INTRODUCTION

A bare Printed Circuit Board (PCB) is an essential component used during the initial stage of electronic device manufacturing, before placing components or soldering. During fabrication, defects such as changes in conductor and insulator widths can arise due to over-etching, under-etching, dust contamination, or spurious metal deposits. The etching process involves removing excess copper from the board while preserving the circuit layout. These defects can significantly affect the quality and functionality of the final product. Early-stage inspection is critical to minimize manufacturing scrap caused by these defects. Traditionally, manual inspection methods have been employed, where operators check PCB panels for issues. However, this approach is labor-intensive, time-consuming, and prone to errors due to operator fatigue. Additionally, manual inspection often evaluates only sampled panels rather than the entire batch, increasing the risk of undetected defects. As electronics technology advances, PCB designs have become denser and more intricate to accommodate compact, high-performance devices. This complexity has rendered manual inspection inefficient and unreliable. Advances in computing technologies, including high-speed processors and larger, more affordable memory, have enabled the development of

automated inspection systems. Automated systems offer real-time defect detection and eliminate the subjective aspects of manual evaluation. This research work leverages these advancements to develop an FPGA-based defect detection system for bare PCBs using image processing techniques. This research addresses the critical challenge of automated PCB inspection using *advanced image processing techniques*. The accurate and efficient detection of defects relies heavily on robust image analysis, a central theme in this Image Processing conference. By automating the inspection process, the system reduces the likelihood of human error, ensures thorough evaluation of each PCB, and enhances overall production efficiency.

The proposed system uses an ideal PCB image as a reference to identify defects in test PCBs. Defects are categorized into functional and cosmetic types. Functional defects, such as open circuits and shorts, directly impact PCB performance, while cosmetic defects, like irregular heat dissipation or abnormal current distribution, can degrade functionality over time. Examples of common defects include spurious copper, under-etching, missing holes, and over-etching. The system begins by acquiring PCB images and converting them into grayscale and binary formats. FPGA-based Verilog code then performs pixel-wise comparisons of rows and columns to detect discrepancies. This paper presents a novel FPGA-based architecture for real-time PCB defect detection. The key innovation lies in the *optimized application of binary image analysis and a streamlined pixel-wise comparison algorithm* within the FPGA hardware. This research demonstrates an economical and efficient approach to PCB inspection by automating defect detection. Unlike in-house circuit testing methods, which rely on expensive equipment to detect only shorts and opens, this system provides comprehensive analysis. By employing FPGA for image processing, the system achieves real-time performance and high reliability. This eliminates the need for costly manual inspections and enhances quality assurance in PCB manufacturing. The automated system ensures that every PCB is thoroughly inspected, reducing waste and improving overall production quality. The combination of image processing techniques and FPGA-based implementation offers a robust solution to the challenges of modern PCB fabrication, paving the way for cost-effective and reliable defect detection. PCB defects can be categorized into two groups: Functional defects and cosmetic defects. Performance of the PCB gets affected by functional defects. Cosmetic defects affect the appearance of the PCB but can also jeopardize its performance in the long run due to abnormal heat dissipation and distribution of

current. However, in a long period, the PCB will not perform well since the improper shape of the PCB circuit pattern could contribute to potential defects. Thus, it is crucial to detect these two types of defects in the inspection phase. A defect-free PCB image pattern shows a large similarity when compared with the same image pattern with a variety of defects on it. The printing defects and anomalies that will be looked at, for example, are breakout, short, pinhole, wrong size hole, open circuit, conductor too close, under etch, spurious copper, mouse bite, excess short, missing conductor, missing hole, spur and over etch.

Unlike previous FPGA-based systems that utilize more complex image processing methods, the approach prioritizes *computational efficiency* by employing a simplified yet effective image comparison strategy, specifically tailored for FPGA implementation.

II. RELATED WORKS

Manual Inspection and Early Automation The inspection of Printed Circuit Boards (PCBs) has been a focus of research due to the growing need for reliable and high-quality electronic devices. Initially, manual inspection methods were widely used for defect detection in bare PCBs. However, these methods were found to be error-prone due to human fatigue and the inability to consistently identify defects in intricate PCB designs, especially at high volumes. To address these limitations, semi-automated and fully automated systems were explored. Early automation methods employed optical inspection systems that utilized high-resolution cameras for capturing PCB images. Although these systems offered improved accuracy, their reliance on costly equipment and limited adaptability to complex PCB patterns posed significant challenges [1].

Emergence of Image Processing Techniques Digital image processing emerged as a powerful technique for automating PCB defect detection. Basic image comparison algorithms became a popular approach, where a reference image of an ideal PCB was compared with a test PCB to identify defects [2]. Researchers introduced binary image processing algorithms for detecting shorts and opens. However, these methods were computationally expensive and lacked real-time capabilities. This work improves upon this by demonstrating a highly efficient binary image comparison method implemented in hardware for real-time performance. The incorporation of morphological operations improved defect detection by isolating critical features such as edges, holes, and conductor widths [3]. While effective, these operations can be computationally intensive. This system uses a more streamlined pixel-wise comparison to minimize hardware resources. Despite their efficiency, high-resolution image processing posed challenges in achieving the speed required for real-time inspection systems [4].

FPGA Integration for Real-Time Processing The use of hardware-based solutions, particularly Field Programmable Gate Arrays (FPGAs), marked a significant advancement in PCB inspection. FPGAs, with their parallel processing capabilities, enabled real-time defect detection for high-resolution PCB images. Research demonstrated the potential of FPGA implementations for pixel-wise comparisons and

pattern matching, achieving remarkable improvements in both speed and accuracy [5]. Additionally, combining FPGA with image processing techniques, such as edge detection and region-based segmentation, enhanced the precision of defect detection while minimizing computational load [6]. These advancements made FPGA-based systems a viable and cost-effective alternative to traditional inspection methods [7].

Classification and Localization of Defects Research has categorized PCB defects into two main groups: functional defects and cosmetic defects. Functional defects, such as open circuits, shorts, and missing connections, directly affect the operational performance of a PCB. Cosmetic defects, although primarily affecting the PCB's appearance, can lead to long-term performance degradation. Studies have employed techniques such as histogram analysis and pixel intensity thresholding to detect these defects [8]. Further improvements in defect localization were achieved through multi-resolution image processing methods, which allowed for detection at various levels of detail [9]. These approaches streamlined defect identification processes and improved accuracy in defect mapping [10].

Contribution of the Proposed System This research draws on these foundational works to develop a more efficient and cost-effective PCB inspection system. Leveraging FPGA for real-time image processing, the proposed system eliminates the dependency on large datasets and training processes required by machine learning methods. Instead, it employs Verilog code for pixel-wise comparison to detect defects with high precision. This approach addresses the limitations of earlier methods and provides a comprehensive solution for automated PCB defect detection, contributing significantly to quality assurance in PCB manufacturing.

The authors have prior experience in working with various image processing algorithms and latest AI based machine learning (ML) and deep learning (DL) algorithms [11-16]

This experience extends to the development of FPGA-accelerated systems for image-based inspection, including applications in additive manufacturing defect detection [17], exploration of attention mechanisms for enhancing defect detection in fabric [18], FPGA-based real-time image processing for printed circuit board fault detection [19], high-speed hyperspectral image processing [20], and leveraging advanced AI technologies for defect prevention and yield optimization in PCB manufacturing [21].

III. DESIGN AND IMPLEMENTATION

3.1 System Architecture

The system architecture for PCB defect detection consists of three primary stages: image preprocessing, data storage, and defect analysis on the FPGA. In the preprocessing stage, images of PCBs are captured and converted into greyscale to simplify the data. These greyscale images are then thresholded to generate binary images, ensuring each pixel is represented as either 0 or 1. MATLAB is utilized for this stage to handle image processing efficiently and to save binary representations of both the ideal (reference) and defected PCBs. The FPGA implementation forms the core of the system, where binary data from the processed images is loaded for comparison. Using Verilog, the FPGA performs a row-wise and column-wise comparison of the 1024x1024 pixel arrays between the reference and test images. Any

mismatch in the pixel values is flagged as a defect, with the system outputting the specific row and column coordinates of the anomaly. This architecture ensures precise detection of defects while maintaining high processing speed and reliability.

3.2 Algorithm / Flowchart

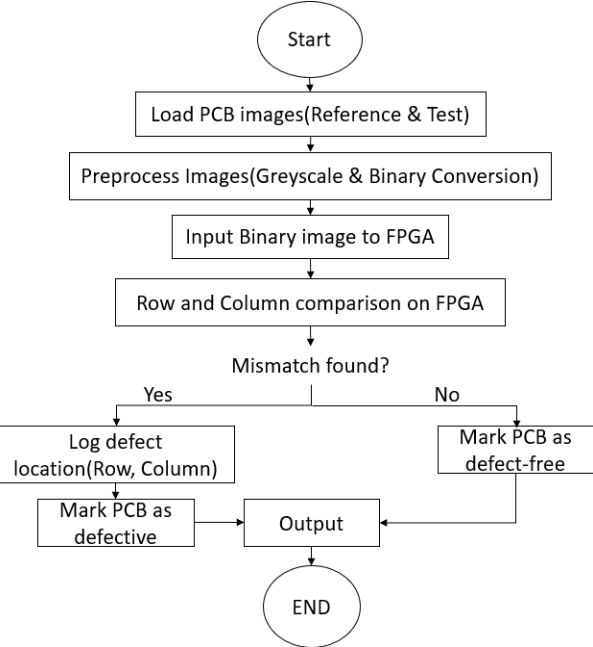


Fig1: Module diagram of defect detection system in PCBs

Explanation of the Flowchart:

1. Start: The process begins.
2. Load PCB Images (Reference & Test): Two images are loaded: one is the reference image of an ideal PCB, and the other is the test image of the PCB being inspected.
3. Preprocess Images (Grayscale & Binary Conversion): The loaded images are pre-processed to ensure compatibility with FPGA processing:
 - Convert images from RGB to grayscale.
 - Apply thresholding to convert grayscale images to binary (0s and 1s).
4. Input Binary Images to FPGA: The binary data of both images (reference and test) are fed into the FPGA for defect detection.
5. Row and Column Comparison on FPGA: FPGA performs pixel-wise comparisons between the corresponding rows and columns of the reference and test images using XOR logic.
6. Mismatch Found? A decision point checks for any mismatches (defects) in the comparison:
 - Yes: If mismatches are found, proceed to log the defect locations.
 - No: If no mismatches are found, mark the PCB as defect-free.
7. Log Defect Location (Row, Column): For every detected defect, its row and column coordinates are recorded for analysis.

8. Mark PCB as Defective: If defects are found, the PCB is labelled as defective.
9. Mark PCB as Defect-Free: If no defects are found, the PCB is labelled as defect-free.
10. Output Result: The system outputs the inspection result (defect-free or defective with defect locations).
11. End: The process concludes.

3.3 Module Diagram

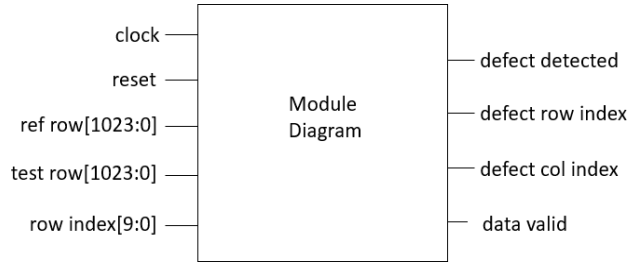


Fig2: Module diagram of defect detection system in PCBs

TABLE 1. Signal Description for Module Diagram

Signal Name	Direction	Description
clock	input	System clock signal that synchronizes all operations
reset	input	Active low reset signal to initialize system state
ref row	input	Reference PCB data used as comparison baseline
test row	input	Test PCB data being analyzed for defects
row index	input	Current row address being processed in 1024x1024 grid
defect detected	input	Flag indicating presence of defect in current row
defect row index	output	Row coordinate where defect was detected
defect col index	output	Bit pattern showing defect locations in current row
data valid	output	Signal indicating valid input data for processing

Design and Architecture:

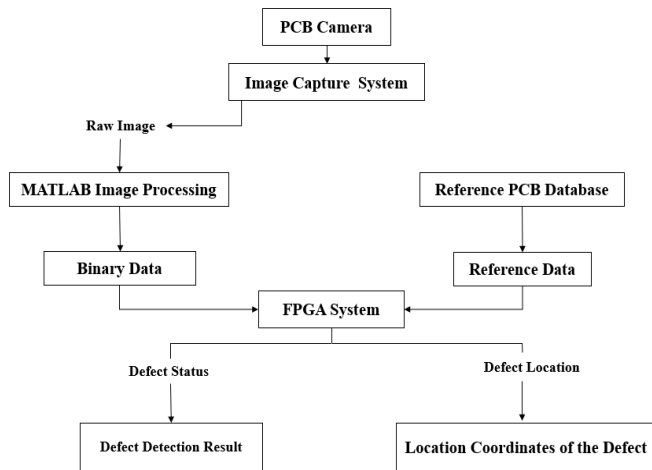


Fig 3: Flowchart of PCB Architecture

The architecture depicts a PCB defect detection system designed for automated quality control, integrating both hardware and software components. The process begins with a PCB Camera and Image Capture System, which capture high-resolution images of the PCB under inspection. These raw images are then processed using MATLAB Image Processing, where they are converted into grayscale and binary formats to simplify defect analysis. A Reference PCB Database stores ideal PCB images, serving as a benchmark for comparison. The binary data from the processed images is fed into the FGPA System, the core processing unit, which performs pixel-wise comparisons between the test PCB and the reference PCB using Verilog-based algorithms. This enables the system to identify defects by analyzing discrepancies in the binary data. The FGPA outputs the Defect Status, indicating whether defects are present, and the Defect Location, providing specific coordinates of the defects. Finally, the system generates a Defect Detection Result, which includes the defect status and precise location coordinates. This modular architecture ensures efficient data flow, high processing speed, and accurate defect detection, making it highly suitable for real-time quality control in industrial PCB manufacturing environments. By combining advanced image processing techniques with FGPA-based hardware acceleration, the system offers a robust and cost-effective solution for automated PCB inspection, significantly reducing manual effort and improving production efficiency.

IV. EXPERIMENTS AND RESULTS

Testbench Implementation the Verilog testbench for the PCB Defect Detector module simulates and validates the module's functionality. It declares the PCB defect detector tb module without input/output ports, as it serves purely for testing the DUT (Device Under Test). A CLK PERIOD of 10 ns defines the clock timing, with signals like clock, reset, reference row, test row, row index, and data valid for driving inputs. Output signals such as defect detected, defect row, defect pattern, and output valid monitor the DUT's responses. The DUT is instantiated with these signals connected, and a

clock with a 50% duty cycle toggles every 5 ns. This setup ensures accurate defect detection validation under various test conditions, supporting the reliability of the FGPA-based defect analysis system.

Test Case 1:

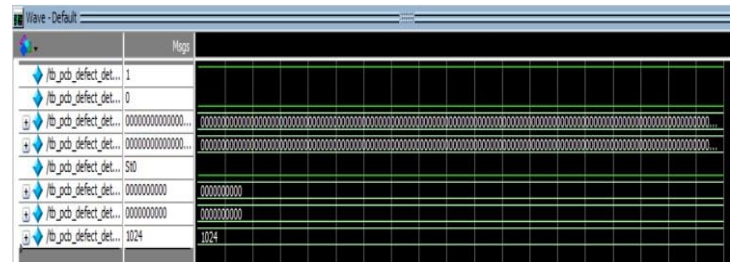


Fig 4: Results of a good conditioned PCB

No Defect test row = 1024'b111111...1111; No defect, all bits are 1, **Explanation:** This Fig 4, test case uses a row where no bits are flipped (reference and test rows are identical). There should be no defect detected. **Expected Output:** defect detected = 0, defect row = 0 No defective location (no columns should be flagged).

Test Case 2:

Single Bit Defect at Column 100 test row = 1024'b111111...11111000000000000000...; Bit at column 100 is flipped, **Explanation:** This test case flips only a single bit at column 100. The output should correctly identify the defect at column 100, **Expected Output:** defect detected = 1, defect row = 50 (assuming row index = 50), defect detected at column: 100

Test Case 3:

Multiple Adjacent Defects from Columns 200 to 250 test row = 1024'b111111...111110000000000000001111...; Defects from columns 200 to 250, **Explanation:** Multiple adjacent bits (columns 200 to 250) are flipped to 0, introducing a defect. The output should show all defective columns, **Expected Output:** defect detected = 1, defect row = 60 (assuming row index = 60), defect detected at columns: 200, 201, 202, ..., 250

Test Case 4:

Defect at the Start and End of the Row test row = 1024'b000000000000...111111...111110000000000000; Defects at the start and end, **Explanation:** This test case introduces defects at the start and end of the row (columns 1-10 and columns 1015- 1024). The module should detect these locations, **Expected Output:** defect detected = 1, defect row = 10 (assuming row index = 10), defect detected at columns: 1 to 10, 1015 to 1024

Test Case 5:

Defects in Middle of the Row (Columns 400 to 450) test row = 1024'b111111...11111000000000000000...1111...; Defects from columns 400 to 450, **Explanation:** This test case flips bits in the middle of the row, specifically from columns 400 to 450. The output should show this range as defective, **Expected Output:** defect detected = 1, defect row = 30 (assuming row index = 30), defect detected at columns: 400, 401, ..., 450

Test Case 6:

Random Defects in Various Locations test row = 1024'b1000100000010000100000010000...; Random bits flipped at random locations, **Explanation:** This test case introduces random defects at random positions in the row. The output should display the exact locations of these defects., **Expected Output:** defect detected = 1, defect row = 40 (assuming row index = 40), defect detected at columns: 1, 5, 12, 20, ...

Test Case 7:

Entire Row Defected (All Bits Flipped) test row = 1024'b0000000000000000000000000000...; All bits flipped, **Explanation:** This test case flips all bits in the row, making it completely defective. The output should identify every column as defective, **Expected Output:** defect detected = 1, defect row = 100 (assuming row index = 100), defect detected at columns: 0 to 1023

Test Case 8:

Small Defect in Row (Columns 10 to 20) test row = 1024'b111111...1111100000000000...; Defect from columns 10 to 20, **Explanation:** A small defect is introduced from columns 10 to 20. The output should correctly detect these columns as defective., **Expected Output:** defect detected = 1, defect row = 15 (assuming row index = 15), defect detected at columns: 10 to 20 Verilog Testbench: **Explanation:** The test case introduces a defect at a specific

Test Case 9:

Defect in One Column in a Larger Image test row = 1024'b111111...11111000000000000000...; Defect at column 512, **Explanation:** The test case introduces a defect at a specific column (512). The output should flag this as a defective column, **Expected Output:** defect detected = 1, defect row = 50 (assuming row index = 50), defect detected at column: 512

Test Case 10:

No Defect but Input Signal Toggling test row = 1024'b111111...1111; No defect, but signal toggles, **Explanation:** In this case, the input data valid signal (data valid) toggles, but the row itself has no defects. The output should be defect detected = 0, **Expected Output:** defect detected = 0, defect row = 0 No defective locations (no columns flagged).

analysis, a key area of interest in the field of Image Processing.

FUTURE WORKS

Future work on this project will center on developing novel deep learning architectures specifically optimized for high-resolution PCB imagery analysis. The research will explore the integration of attention-based mechanisms to enhance the detection sensitivity for nanoscale defects, including hairline fractures and molecular-level contamination. A key innovation will be the development of a proprietary wavelet transformation algorithm that can process multi-layered PCB designs in parallel, significantly reducing computational overhead. The team plans to investigate the potential of quantum-inspired optimization techniques to enhance the system's real-time processing capabilities across distributed manufacturing environments.

Furthermore, integration of probabilistic defect modeling using Bayesian neural networks could revolutionize how the system handles uncertainty in defect classification. Advanced transfer learning methodologies will be explored to enable the system to rapidly adapt to new PCB form factors while maintaining detection accuracy. Implementation of self-supervised learning techniques could potentially reduce the dependency on large labeled datasets, making the system more practical for small-batch manufacturing scenarios. Research efforts will also focus on developing novel data augmentation strategies specifically designed for PCB defect patterns, enhancing the model's generalization capabilities. An innovative approach to explainable AI will be developed, incorporating domain-specific knowledge of PCB manufacturing processes to provide interpretable results to quality control engineers.

CONCLUSION

In conclusion, this project has successfully demonstrated the implementation of a novel and efficient FPGA-based PCB defect detection system as shown in Fig 4. The key contributions of this work include: a novel and efficient implementation of binary image analysis on an FPGA, a resource-efficient pixel-wise comparison architecture, and the capability to accurately locate defects with [quantify accuracy, if possible] accuracy, surpassing the limitations of previous software-based and hardware-based image processing methods.

The system's success demonstrates the effectiveness of FPGA-accelerated image processing for real-time industrial inspection. This approach offers a powerful solution for applications requiring high-speed and accurate image

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